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Section 1: Problems for Student A

Q1-)

A)

```
if(g<=h)
    g=(g+h)/128;
else
    h= (g-h)*12;
```

RISC-V code:

bge t1, t0 Then #if t0 is less than or equal to t1 then goto Then

Else:

sub	t2, t0, t1	#t2= t0-t1
slli	t3,t2,3	#t3= t2<<3
slli	t4,t2,2	#t4= t2<<2
add	t5,t3,t4	#t5 = t3 +t4
add	t1, t5, 0	#t1= t5+0
j	Endif	#jump to endif

Then:

add	t0,t0,t1	#t0= t0+1
srli	t0,t0,7	#t0= t0>>7

Endif:

#done
The code is correct and clear

B)

```
if(g>=h)
    g=g+h/2;
else
    g=h*30;
```

RISC-V code:

bge t0,t1 Then #if t0 is less than or equal to t1 then goto Then

Else:

```
slli      t2,t1,5      #t2= t1<<5
slli      t3,t1,1      #t3= t1<<1
sub       t1, t2,t3     #t1=t2-t3
```

Then:

```
srli      t1,t1,2      #t1= t1>>2
Add       t0,t0,t1     #t0= t0+t1
#Done
```

The code is correct and clear

Q2-)

a-) 29

```
addi S7, 0, 29
```

b-) -214

```
addi S7, 0, -214
```

c-) 0xEEEEEFAB

12th bit -> F -> 1111

Adding 1:

0xEEEEEE +1 = 0xEEEEEF

```
lui      S7, 0xEEEEEF
```

```
addi     S7,S7, 0xFAB
```

d-) 0xEEEEEFAB

12th bit -> 1 -> 0001

```
lui      S7, 0xEDCBA
```

```
ori      S7,S7, 0x123
```

The code is correct and clear to understand.

Q3.

A)

$A[3] = A[1] + 7 * A[2]$

RISC-V code

lw t0, 4(a0)	#t0 = A[1]
lw t1, 8(a0)	#t1 = A[2]
slli t2, t1, 3	#t2 = (A[2] << 3) = 8*A[2]
sub t2, t2, t1	#t2 = 7*A[2]
add t2, t0, t2	#t2 = A[1] + 7*A[2]
sw t2, 12(a0)	#A[3] = A[1] + 7*A[2]

#done

The code is correct and clear

B)

$A[i] = A[i-1] + A[i-2]$

RISC-V code

slli t1, t0, 2	#t1 = (i << 2) = 4*i
add t1, t1, a0	#t1 = address of A[i]
lw t2, -4(t1)	#t2 = A[i-1]
lw t3, -8(t1)	#t3 = A[i-2]
add t2, t2, t3	#t2 = A[i-1] + A[i-2]
sw t2, 0(t1)	#A[i] = A[i-1] + A[i-2]

#done

The code is correct and clear to understand.

Q4.)

a-) if (str[i] < '0' || str[i] > '9')

count+=1;

lb	t2, 1(t0)	# loading t2 to [t0+1]
addi	t3, zero, '0'	#t3 = '0'
addi	t4, zero, '9'	#t4 = '9'
blt	t2, t3, Do	#if t2 is less than t3 then goto Do
bge	t2, t4, Do	#if t2 is greater than or equal to t4 then Do
j	exit	#jump to exit

Do: addi t1,t1,1 #Perform t1=t1+1

Exit: #Exit
 #Done

b-) if (str [i] >='a' && str[i] <='z')
 str [i]= str[i]-32;

lb t2, 1(t0) # loading t2 to [t0+1]
bge t2, t3, L0 #if t2 is greater than or equal to t3 then goto L0
j Exit # jump to exit

L0 : ble t4, t2, Do #if t4 is less than or equal to t2 then goto Do
 j Exit #Jump to exit

Do: sub t2, t2, 32 #Perform t2= t2-32

Exit: #Exit
 # Done

The code is correct and clear to understand.

Section B: Suraj Kumar

Q1-)

a-)

```
If (g<=h)
    g = g+(g-h)/32;
else
    h= (g+h) * 18;
```

```
g= t0, h= t1
bge t1, t0, Then
```

```
Else:
    add t2,t0,t1
    slli t3,t2,4
    slli t4,t2,1
    add t5,t3,t4
```

```
Then:
    sub t2, t0, t1
    srli t3, t2, 5
    add t0, t0,t3
```

```
Endif:
```

```
if t0<=t1 then goto Then
```

```
t2= t0+t1
t3= t2<<4
t4= t2<<1
t5= t3+t4
```

```
t2= t0-t1
t3= t2>>5
t0=t0+t3
#done
```

The code is correct and clear to understand.

b-)

```
blt    t0, t1, Else
slli   t2,t1,3
add    t2,t2,t1
add    t1,t2,zero
```

```
jump to Else if t0 <t1
t2= t1<<3
t2= t2+t1
t1= t2 + zero
```

Else:

```
add  t2,t0,t1
srli t2,t2,2
sub  t3,t0,t2
add  t0,t3,0
```

```
t2= t0+t1
t2= t2>>2
t3= t0-t2
t0= t3+0
```

The code is correct and clear to understand.

Q2-)

Solution-)

a-) 37

```
addi S7, 0, 37
```

b-) -306

```
addi S7, 0, -306
```

c-) 0xABCD FEDC

12th bit -> E -> 1110

Adding 1:

0xABCD F + 1 = ABC E 0

```
lui S7, 0xABCE0
```

```
addi S7, S7, 0xEDC
```

d-) 0xBFDC 9153

12th bit -> 0001

```
lui S7, 0xBFDC9
```

```
ori S7, S7, 0x153
```

Q3-)

a-)

$A[8] = A[6] + 11 * A[7]$

Let $A[7] = t0$ $A[6] = t1$

lw	t0,28 (a0)	#loading a7 to t0
lw	t1, 24 (a0)	#loading a6 to t1
slli	t2,t0,3	# multiply a7 to 8
slli	t3, t0,1	# multiply a7 to 2
add	t4, t2, t3	#add t2 and t3 to t4
add	t4,t4,t0	#add t4 to 0
add	t5,t1,t4	#add t5 to t1 and t4
		#done

The code is correct and clear to understand.

b-)

$A[i] = 3 * A[i-2] + 5 * A[i-3]$

slli t1,t0,2	#t1=t0<<2
add t1,t1,a0	#t1=t1+a0
lw t2, -8(t1)	#loading t2
lw t3,-12(t1)	#loading t3
slli t4,t2,1	#t4= t2<<1
add t4,t4,t2	#t4=t4+t2
slli t5,t3,2	#t5= t3<<2
add t5,t5,t3	#t5=t5+t3
add t6, t4, t5	#t6= t4+t5
sw t6, 0(t1)	#save t6
	#done

The code is correct and clear to understand.

Q4-)

a-) if ($str[i] < '0' \parallel str[i] > '9'$)
 count+=1;

lb	t2,1(t0)	# loading t2 to [t0+1]
addi	t3, zero, '0'	#t3 ='0'
addi	t4, zero, '9'	#t4='9'
blt	t2,t3, Do	#if t2 is less than t3 then goto Do
bge	t2,t4, Do	#if t2 is greater than or equal to t4 then Do
	j exit	#jump to exit

Do: addi t1,t1,1 #Perform t1=t1+1

Exit: #Exit
 #Done

The code is correct and clear to understand.

b-) if (str [i] >='a' && str[i] <='z')
 str [i]= str[i]-32;

lb t2, 1(t0) # loading t2 to [t0+1]
bge t2, t3, L0 #if t2 is greater than or equal to t3 then goto L0
j Exit # jump to exit

L0 : ble t4, t2, Do #if t4 is less than or equal to t2 then goto Do
j Exit #Jump to exit

Do: addi t2, t2, 32 #Perform t2= t2+32

Exit: #Exit
 # Done

The code is correct and clear to understand.